

Statistical Certification of Autonomous Vehicle Fallback Triggers: Closed-Loop Evaluation Under Dataset Shift

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Abstract—A fallback trigger decides when an automated vehicle should abandon its task for a minimum-risk manoeuvre (MRM). In practice it is a hand-tuned threshold with no stated guarantee. Statistical certification of alert thresholds exists, but the consequences of the fallback that the alert triggers are rarely priced. We ask how reliable a certified trigger is when the fallback is actually executed, and what happens to it under dataset shift. Because a rollout depends only on the tick at which the fallback fires, one reference run and one forced-firing rollout per decision tick evaluate every trigger, threshold, online method and latency exactly, including induced collisions and harm that persists after the fallback fires. On replayed Argoverse 2 and nuScenes logs with reactive traffic, a UN R157 manoeuvre and nuPlan harm labels, we compare a Learn-then-Test-certified geometric trigger with seven baselines and a class-conditional certificate in the style of Luo *et al.* At a 5 % target, the certified trigger misses 3.7 % of scenarios and exceeds the target in 9.5 % of 200 resplits (limit 10 %); the same score tuned without a certificate exceeds it in 43.5 %, at 2.5 points fewer unnecessary stops, so we do not claim a better trade-off. The certificate controls the alert, not the outcome: harm in the intervened run is 6.5 %, including 1.1 % created by the intervention, and a 5 % marginal miss target leaves 24 % of harmful scenarios unalerted. A class-conditional 5 % target degenerates to always firing with this predictor. Confidence and ensemble triggers need 68–71 % unnecessary stops. A threshold certified on Argoverse 2 misses 13.2 % on nuScenes, 5.0 % of nuScenes scenarios contain harm no trigger can prevent, and one decision tick (0.5s) of latency breaks the target. [Waymo replication to be inserted.]

Index Terms—Autonomous driving, conformal prediction, selective prediction, minimum-risk manoeuvre, closed-loop safety evaluation, trajectory prediction, risk calibration, distribution shift.

I. INTRODUCTION

AUTONOMOUS vehicles that lack confidence in their own situational understanding are expected to fall back on a minimum-risk manoeuvre: slow to a stop, pull to the shoulder, or otherwise hand control back to a human. The decision of *when* to invoke this behaviour is, in nearly every deployed and published system, a hand-tuned threshold on

some scalar confidence or uncertainty score produced by a perception or prediction module. That practice has a specific and largely unexamined weakness: the very quantity the threshold is applied to — a learned model’s uncertainty estimate — is exactly what has been shown to degrade under the kind of distribution shift a vehicle experiences when it enters a new city, a new weather condition, or simply a stretch of road unlike anything in its calibration set. A threshold tuned under one operating distribution carries no guarantee once that distribution has moved, and the failure is silent: the system continues to report a confidence number, and nothing in the number itself signals that the number is now wrong.

Conformal prediction offers a principled alternative: rather than a point confidence score, it produces a prediction region with a distribution-free, finite-sample coverage guarantee under exchangeability. A substantial and fast-growing literature has applied conformal prediction to trajectory forecasting and to planning under uncertainty in autonomous driving. Almost without exception, however, this literature is *open-loop*: it constructs a prediction region, reports its coverage and its size, and stops. Two things are invisible from that vantage point. First, a region’s effect on a downstream control decision — specifically, whether it is used to trigger a fallback — is not evaluated at all; open-loop coverage and closed-loop safety outcome are simply assumed to move together. Second, and more consequentially, *triggering the fallback is not free*. A vehicle that stops in a live lane because its uncertainty region grew large is not obviously safer than one that continues; an evaluation that reports only coverage, or only collision rate under the fallback, cannot see the trade-off between the two, and cannot see that a policy which triggers whenever it is asked to would trivially achieve a good miss rate while being useless.

This paper asks a narrower question than “which trigger is best”: *how reliable is a statistically certified fallback trigger when it is judged through the interventions it actually causes, and what happens to it when the deployment data differ from the calibration data?* We treat the trigger as a decision problem with a formal guarantee attached to the decision itself, and evaluate it where that guarantee has to hold: in closed loop, on real driving scenarios, with a threshold whose consequences change what the vehicle does next. Certifying a threshold on a risk score against missed events is not new; Luo *et al.* [8] do so with conformal prediction, and Conformal Decision Theory [12] calibrates decisions rather than predictions through a long-run *average* guarantee. We

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build on these and do not claim the certification step as ours. An MRM trigger nevertheless differs in kind from the settings they study. It is an irreversible, one-shot decision to abandon the task, its errors are paid in two currencies (missed interventions, and the harm of stopping itself), and a deployed vehicle needs its guarantee before the first mile.

We organise the study around four research questions:

- **RQ1.** Does a certificate keep its promise when measured on the closed-loop outcome, and what does keeping it cost in unnecessary stops and fallback-induced collisions?
- **RQ2.** Do open-loop failure-detection metrics choose the same trigger that closed-loop safety does?
- **RQ3.** What happens to a certificate under a real cross-dataset shift, and how much can be repaired with unlabelled or with a few labelled target scenarios?
- **RQ4.** How sensitive is the certificate to the timing of the intervention?

We make four contributions.

- 1) **An exact evaluation protocol for executed fallbacks.** A rollout's outcome depends only on *when* the fallback fires (Proposition 1). One reference run that records every trigger score, plus one rollout forced to fire at each decision tick, evaluates every trigger, threshold, online method and latency on identical physics. The alert-level quantities (miss, unnecessary stop) need only the reference run. The forced rollouts add the consequences of the fallback itself: induced collisions, route completion, and harm that persists or arises after it fires.
- 2) **What certification buys and what it costs.** Certification is the only design that reliably keeps its target across resplits, at a measured cost in stops. It does not win the safety-conservatism trade-off against a tuned threshold, a hypothesis we pre-registered and report as refuted. The certified quantity is the alert, not the outcome: residual harm after the fallback is larger than the certified miss rate, part of it created by the intervention. Confidence and ensemble triggers, the deployed practice, need several times more unnecessary stops than a geometric score. A class-conditional certificate in the style of Luo *et al.*, implemented in the same framework, is vacuous at a 5 % target on this predictor.
- 3) **The certificate under real dataset shift.** An Argoverse 2 certificate fails on nuScenes. The nuScenes data also expose a floor: harm present at the first decision tick, which no trigger can prevent and which can make a target unreachable. A density-ratio-weighted certificate that reports its effective sample size repairs part of the gap, and in an exploratory analysis the transfer is asymmetric.
- 4) **Operating-point findings for practitioners.** Latency of one decision tick breaks the target, route-following ego corridors make a trigger fire in every scenario, and open-loop AUROC ranks triggers only approximately as closed-loop stops do.

All hypotheses, thresholds and refutation rules were recorded in a version-controlled note before the data were collected. Refuted and unsupported hypotheses are reported

as such, and analyses run after the pre-registered verdicts are labelled exploratory.

The remainder of the paper is organised as follows. Section II situates this work against open-loop conformal prediction for driving, task-relevant failure detection, and closed-loop safety evaluation. Section III formalises the triggering problem and the certification procedure. Section IV describes the closed-loop evaluation environment, the fallback manoeuvre, the harm labels and the baselines. Section V reports results, ablations and the shift experiments. Section VI states the scope of the claims, and Section VII concludes.

II. RELATED WORK

A. Conformal prediction for autonomous driving

Conformal prediction [1] constructs prediction sets with a distribution-free, finite-sample marginal coverage guarantee under exchangeability between calibration and test data, and has been applied extensively to trajectory forecasting and to safe planning under uncertainty. Lindemann *et al.* embed conformal prediction regions directly into a model predictive controller to obtain probabilistic safety guarantees, but their guarantee collapses once the deployment distribution shifts away from the calibration distribution — a limitation Rahaman *et al.* [4] address with robust conformal prediction driven by a conditional generative model of the shift, evaluated in a synthetic multi-agent corridor. SafePath [5] integrates conformal prediction into an LLM-based path-planning pipeline, providing a guarantee that a set of candidate paths contains a safe one with a user-specified probability, and reports both open-loop (nuScenes) and closed-loop (highway-env, synthetic traffic) results. In all three cases the guarantee is on the *prediction* — a region contains the truth, or a path set contains a safe path — rather than on the eventual control decision the region is used to make, and none evaluates a fallback-triggering decision on real, replayed driving logs under reactive traffic. This paper's guarantee is instead on the decision: the probability that a minimum-risk manoeuvre fires before a harmful event, evaluated end-to-end in closed loop.

B. Online and decision-level conformal methods

Adaptive Conformal Inference (ACI) [13] updates the mis-coverage level online so that long-run coverage holds even under arbitrary shift. Conformal Decision Theory (CDT) [12] carries the same idea from prediction sets to decisions: it updates a decision parameter $\lambda_{t+1} = \lambda_t + \eta(\varepsilon - \ell_t)$ so that the long-run average risk converges to ε at rate $O(1/t)$, with no distributional assumptions. It is demonstrated on robot navigation among pedestrians (Stanford Drone Dataset), with a continuous distance-based loss. Both give *asymptotic, average* guarantees for a quantity adjusted continuously as data arrive. We target a different object: a one-shot, irreversible trigger whose certificate must hold with finite samples before deployment, and whose errors include the harm that triggering itself causes. We adopt both methods as baselines (Section IV-E), applied episode-by-episode, which is the natural online counterpart of our setting.

TABLE I
LUO *et al.* [8] VERSUS THIS STUDY.

	Luo <i>et al.</i>	This paper
Guarantee	$\Pr[\text{no alert} \mid \text{unsafe}] \leq \epsilon$, class-conditional	$\Pr[\text{harm not pre-empted}] \leq \alpha$ over scenarios, high probability
Unsafe label	safety score of logged future $< f_0$	nuPlan harm in the no-fallback run
Predictor	Trajectron++	AutoBot
Alert consequence	none (scored on logs)	MRM executed in closed loop
Cost of alerting	false-positive rate	unnecessary stops, induced collisions, residual harm
Data	nuScenes, Lyft	Argoverse 2, nuScenes [Waymo]
Shift	label shift only	dataset shift, weighted repair
Latency	not studied	studied

C. Task-relevant and closed-loop failure detection

Farid *et al.* [6] propose a run-time monitor for trajectory-prediction failures that is *task-relevant* by construction: it flags a misprediction only when the resulting planning cost is anomalous relative to the predicted cost distribution, with data-free false-positive and false-negative rate bounds. Their detector is evaluated against hand-labelled scenarios in an open-loop and semi-reactive setting, and — like the conformal-prediction literature above — does not close the loop through an actuated fallback and re-measure the outcome. Their central insight, that a task-irrelevant misprediction should not count against a detector, is reflected in this paper’s harm labelling: a collision the ego vehicle suffers while already stopped or while being struck from behind is scored as a distinct *induced-collision* cost of the fallback itself, never as a missed intervention.

Closest to our certification step, Luo *et al.* [8] tune the threshold of a warning system with Mondrian conformal prediction so that, among situations that are unsafe in the true future *in the absence of corrective action*, fewer than ϵ raise no alert, using as few as $1/\epsilon$ unsafe examples. Their unsafe label is a threshold on a safety score of the logged future, their predictor is Trajectron++, and their driving experiments (nuScenes and Lyft) score alerts against the logged data; the alert triggers nothing. We adopt the same principle and do not claim it as new. Table I lists the differences that matter here. Their target is class-conditional (a miss rate among unsafe situations), ours is marginal over scenarios; their false-positive rate counts alerts on safe situations, whereas we execute the fallback and price it in unnecessary stops, induced collisions and residual harm; and they study robustness to a change in the proportion of unsafe examples but not to a change of dataset. We implement their certificate inside our framework and report it as a baseline (Section V-E).

Sinha *et al.* [19] close the loop differently: a runtime monitor calibrated with conformal prediction detects when perception leaves the tolerance of a robust MPC controller that keeps a perception-free fallback plan feasible, demonstrated on aircraft taxiing. Their fallback is part of the controller design, whereas ours is an irreversible manoeuvre whose trigger is the object of study.

Scenario-based simulation is a standard route to evaluating automated vehicle safety [20], [21]; we use replay of real logs

with reactive traffic rather than synthetic scenarios.

D. Risk-sensitive safety analysis

Chapman *et al.* [9] develop risk-sensitive safe sets using Conditional Value-at-Risk to assess the severity, not merely the probability, of constraint violation along a stochastic trajectory, providing the risk-sensitive vocabulary this paper borrows in motivating why a mean missed-intervention rate is the appropriate quantity to control at a chosen operating point, even though a tail-severity extension (Section VI) is left to future work.

E. Closed-loop simulation infrastructure

ScenarioNet [10] unifies heterogeneous real-world driving logs — including Argoverse 2, nuScenes, nuPlan, and Lyft Level 5 — into a common scenario description that can be replayed interactively in the MetaDrive simulator, with either logged-trajectory replay or reactive rule-based (IDM) traffic. This work is built directly on that infrastructure: every scenario reported in Section V is a real logged driving scenario, replayed headless with reactive traffic, not a synthetic or hand-crafted one.

III. PROBLEM FORMULATION AND METHOD

A. Fallback triggering as selective prediction

Consider a closed-loop rollout of a scenario X under a fallback policy governed by a scalar threshold λ : at every decision tick the vehicle computes an uncertainty score $u_t(X)$ from a trajectory predictor’s output for nearby agents, and engages the minimum-risk manoeuvre the first time $u_t(X) > \lambda$, remaining engaged for the rest of the episode. Define the closed-loop loss

$$L(X, \lambda) = \mathbb{I}[\text{harm in } X \text{ under } \lambda, \text{ MRM not first}], \quad (1)$$

the *missed-intervention* indicator, and the complementary over-conservatism quantity

$$S(X, \lambda) = \mathbb{I}[\text{MRM fires under } \lambda, X \text{ harm-free at } \lambda = \infty], \quad (2)$$

the *unnecessary stop* indicator, evaluated against the counterfactual rollout of the same scenario with the trigger disabled. The goal is to choose $\hat{\lambda}$ such that

$$\Pr_{X_1, \dots, X_n} \left(\mathbb{E}_X [L(X, \hat{\lambda})] \leq \alpha \right) \geq 1 - \delta, \quad (3)$$

while minimising $\mathbb{E}_X [S(X, \hat{\lambda})]$ — the smallest autonomy loss compatible with the risk target α .

Let $h(X)$ be the first harm step of the reference run with the fallback disabled (∞ if none) and $\tau_\lambda(X)$ the firing step. Because a run under the trigger coincides with the reference run until it fires (Section III-E), $L(X, \lambda) = \mathbb{I}[h(X) \leq \tau_\lambda(X)]$ and $S(X, \lambda) = \mathbb{I}[\tau_\lambda(X) < \infty, h(X) = \infty]$: both alert-level indicators depend only on the reference run and the firing step. The consequences of the fallback after it fires enter through separate quantities (induced collisions, residual harm, route completion), which we report but do not certify. Since the running maximum of the score is non-decreasing, τ_λ is non-decreasing in λ and $L(X, \lambda)$ is non-decreasing in

λ for every scenario. Equation (3) is therefore a monotone-risk statement, to which Conformal Risk Control [3] applies directly, but with an expectation guarantee: it bounds the calibration-averaged risk, not the probability that a given calibrated threshold exceeds α . We use Learn-then-Test [2] for the high-probability form: for a finite grid $\lambda_1 > \lambda_2 > \dots > \lambda_J$ ordered from least to most conservative, compute the empirical risk $\hat{R}(\lambda_j) = \frac{1}{n} \sum_i L(X_i, \lambda_j)$ on n calibration scenarios, a Hoeffding–Bentkus p -value for the null hypothesis $H_j : \mathbb{E}[L(\cdot, \lambda_j)] > \alpha$, and apply fixed-sequence testing starting from the most conservative λ ; every λ_j whose null is rejected satisfies (3) simultaneously, with family-wise error controlled at δ , and $\hat{\lambda}$ is the least conservative such λ_j (largest autonomy). We report the violation frequency of both procedures over resplits rather than assuming it.

The marginal target is weaker than it may sound. With harm in a fraction π of scenarios, a marginal miss probability α leaves a fraction α/π of harmful scenarios unalerted, which is 24% at $\alpha = 0.05$ and $\pi = 0.158$ in our data (Section V-E).

B. Trigger score

At each decision tick the trigger score u_t measures how far a predicted future position of any nearby agent intrudes into the ego vehicle’s own near-term extent. Writing the ego’s constant-heading, constant-speed extrapolation over a horizon of T steps as $\{p_\tau^{\text{ego}}\}_{\tau=1}^T$ — the same route-agnostic motion model used by the harm metric in Section IV-B, deliberately *not* the vehicle’s planned, route-following path, so that the trigger cannot credit itself for steering behaviour the safety label does not assume — and a predictor’s K -mode forecast for agent i as $\{\hat{p}_\tau^{(i,k)}\}$, each predicted offset is decomposed into a longitudinal component $a_\tau^{(i,k)}$ (along the ego heading) and a lateral component $\ell_\tau^{(i,k)}$ (perpendicular to it), each compared against a separate margin: a length-based longitudinal margin $m_i^{\text{lon}} = \frac{1}{2}(L_{\text{ego}} + L_i) + c_{\text{lon}}$ and a width-based lateral margin $m_i^{\text{lat}} = \frac{1}{2}(W_{\text{ego}} + W_i) + c_{\text{lat}}$. The intrusion at (i, k, τ) is

$$\rho_\tau^{(i,k)} = \max \left(0, 1 - \sqrt{\left(\frac{a_\tau^{(i,k)}}{m_i^{\text{lon}}} \right)^2 + \left(\frac{\ell_\tau^{(i,k)}}{m_i^{\text{lat}}} \right)^2} \right) m_i^{\text{lat}}, \quad (4)$$

and $u_t = \max_{i,k,\tau} \rho_\tau^{(i,k)}$. A single isotropic clearance radius cannot represent both margins at once: a radius set by width alone risks missing a straight-ahead lead vehicle, and one set by the half-diagonal risks triggering on adjacent-lane traffic. The decomposition in (4) is therefore a modelling choice made during development on this reasoning. Our ablation (Section V-C) finds that it gives no measurable benefit over a width-only isotropic radius, so we report it as a design option rather than a requirement.

C. Fallback manoeuvre

On triggering, the vehicle executes the minimum-risk manoeuvre defined by UN Regulation No. 157 (Automated Lane Keeping Systems) §5.5.1 [18]: decelerate inside the current lane with a deceleration demand not exceeding 4.0 m/s^2 , lane-keeping control otherwise unchanged, holding the vehicle

stationary once stopped. This is enforced by a proportional controller tracking the target speed profile $v(t) = \max(v_0 - 4.0t, 0)$ from the moment of triggering.

D. Certification under covariate shift

The certificate in (3) assumes calibration and deployment scenarios are exchangeable. When the deployment distribution differs by covariate shift, we reweight calibration scenarios by an estimated density ratio $w(x) \propto p_{\text{target}}(x)/p_{\text{source}}(x)$ computed from pre-deployment scenario covariates only (agent counts and speeds, initial gap, route length). A probabilistic domain classifier is trained on unlabelled target scenarios, following weighted conformal prediction [14]. The weighted empirical risk $\hat{R}_w(\lambda) = \sum_i w_i L(X_i, \lambda) / \sum_i w_i$ is tested with the Hoeffding–Bentkus p -value at the Kish effective sample size $n_{\text{eff}} = (\sum_i w_i)^2 / \sum_i w_i^2$. As the weights concentrate, n_{eff} shrinks and the test certifies a more conservative threshold, and eventually none at all. The procedure thus reports when the source data cannot support a guarantee for the target, rather than asserting one that has silently expired.

E. Exact evaluation of every trigger

Fix a scenario X and its random seed. The *reference run* $R(X)$ disables the fallback. For a decision tick j , the *forced run* $F(X, j)$ is identical to $R(X)$ until tick j and then executes the MRM for the rest of the episode. Let $\Phi(X, j)$ denote any outcome of $F(X, j)$ (miss, stop, induced collision, residual harm, route completion) and $\Phi(X, \infty)$ the outcome of $R(X)$.

Proposition 1 (Firing-tick sufficiency): Assume (A1) the simulator, background traffic and predictor are deterministic functions of the seed and of the ego’s trajectory, so two runs with the same seed and the same ego control up to a step have identical states up to that step; (A2) the trigger’s decision at tick j depends only on the state history up to tick j ; (A3) once the fallback fires it is absorbing, and the control after firing depends only on the state at firing. Then for every trigger with firing tick $\tau(X) \in \{0, \Delta, 2\Delta, \dots\} \cup \{\infty\}$, the run under the trigger is $F(X, \tau(X))$, and every outcome equals $\Phi(X, \tau(X))$. This holds for any score, threshold, online update rule and actuation latency ℓ (for which the run is $F(X, \tau(X) + \ell)$).

Proof: By induction over steps. Until τ the trigger has not fired, so by (A3) the control is that of $R(X)$ and by (A1) the states equal those of $R(X)$; by (A2) the trigger’s decision at each such tick is a function of these states, hence the same in every run sharing them, so τ is determined by $R(X)$ alone. At τ the fallback engages with a control that depends only on the state at τ (A3), which is the control of $F(X, \tau)$. Both runs then have equal states by (A1). ■

Two corollaries follow. First, alert-level indicators need no forced run: L and S are functions of $R(X)$ and τ alone. Second, the consequences of the fallback need n_{ticks} forced rollouts per scenario, after which any number of triggers is evaluated by lookup, whereas direct execution needs one rollout per trigger and setting. Assumption (A1) is an idealisation. We measure its violation in every forced run: over the first 33,761 forced rollouts (2,000 scenarios), 98.4% agreed with the reference run to within 1 cm before firing, the pre-fire

time-to-collision trace differed in 1.2 %, and the pre-fire harm status in one rollout (0.003 %). We report every analysis on all scenarios and after dropping the 3.6 % of scenarios with any flagged drift. All methods are compared on the same simulated physics, so differences between them cannot come from simulator noise or unequal compute.

IV. EXPERIMENTAL SETUP

A. Simulation environment

All rollouts are executed headless in MetaDrive, replaying scenarios converted by ScenarioNet [10] from Argoverse 2 Motion Forecasting and nuScenes. Non-ego traffic is controlled reactively (rule-based, IDM), not replayed open-loop from the log: pure log replay was found to roughly triple the no-fallback collision rate on a 1,000-scenario diagnostic sample, because logged agents cannot react to an ego vehicle that deviates from its logged trajectory — precisely the deviation a fallback manoeuvre produces. Reporting results under non-reactive replay would therefore misrepresent the cost of stopping, which is a central quantity this paper reports.

B. Harm labelling

A rollout is labelled harmful using the nuPlan closed-loop safety definition [11]: an at-fault collision, or a time-to-collision below 0.95 s computed over a 3.0 s horizon at 0.1 s resolution, not evaluated while the ego is stationary and not evaluated against tracks behind the ego. Fault attribution follows the nuPlan convention: a collision the ego suffers while stationary, or while being struck from behind, is *not* at-fault and is never counted as a missed intervention; it is instead recorded as an *induced collision* — the concrete, measurable cost of a false or premature trigger.

C. Predictor

The trajectory predictor is AutoBot [16] (6 modes, 2.1 s history, 6 s horizon), trained with the UniTraj framework [17] on the Argoverse 2 Motion Forecasting training split (minADE₆ = 1.09 m on held-out AV2 validation scenes). It runs online inside the closed loop, so its input is the actual, possibly ego-deviated, simulator history rather than the original log. This is necessary once traffic is reactive. The model’s own UniTraj preprocessing is reused unchanged, so it sees its training input distribution. Vehicles are predicted by the network. Pedestrians and cyclists, which the checkpoint was not trained on, receive constant-velocity forecasts.

D. Datasets and splits

All scenarios come from Argoverse 2 validation, disjoint from the predictor’s training data. They are split deterministically by scenario identifier into development (14,990), calibration (4,971) and test (5,027) sets. The shifted target is nuScenes (9,041 scenarios of 8.1 s, Boston and Singapore, versus Argoverse 2’s six U.S. cities). Scenarios in which the ego moves less than 10 m are excluded, as in ScenarioNet. Design choices were made on the development split only. The pre-registered evaluation uses 2,000 calibration, 2,000

test and 1,500 nuScenes scenarios. Ten Argoverse 2 scenarios (six calibration, four test) whose per-agent records could not be matched after a re-scoring pass are dropped from every analysis, so all methods are compared on identical scenarios; this leaves 1,994 calibration and 1,996 test scenarios, and 1,504 nuScenes scenarios. A second shifted target, Waymo Open Motion [7] validation scenarios (9.1 s, 1 s of history), is collected under a separate pre-registered amendment [results pending]. Decisions are taken at 2 Hz. A paired development-split comparison showed identical miss rates at 2 Hz and 10 Hz, and a 10 Hz replication on 100 test scenarios is reported.

E. Baselines

All methods use the same predictor, the same MRM and the same test scenarios, and are evaluated exactly (Section III-E).

- **T1 — confidence threshold** (deployed practice): fire when $1 - \max_k p_k$ of the network-predicted vehicle whose predicted modes come closest to the ego exceeds a threshold. Agents the network does not predict (pedestrians, cyclists, distant vehicles) are excluded from this score. The threshold is tuned on calibration data to the most permissive value with empirical miss rate $\leq \alpha$. A variant that uses the nearest agent ahead is also evaluated.
- **T2 — ensemble threshold**: the same tuning, applied to the spread of endpoint predictions across five Monte Carlo dropout passes [15].
- **T3 — open-loop conformal**: inflate every predicted mode by the split-conformal radius of the predictor’s 3 s error on held-out Argoverse 2 logs at level α [1], and fire when the inflated region touches the ego. This is the guarantee-on-the-prediction design that the paper argues is insufficient.
- **Tuned geometric**: our score with an empirically tuned threshold and no certificate. It isolates the contribution of the calibrator.
- **CDT** [12] and **ACI** [13]: online decision- and prediction-level adaptation, updated after every scenario. Each uses the step size that performs best on test, which favours the baselines.
- **Oracle**: the best threshold in hindsight on test data. It is not deployable and is reported as an upper bound.

F. Metrics and statistics

For every method we report the missed-intervention, unnecessary-stop and fallback-induced-collision rates, autonomy (the fraction of scenarios completed without the fallback) and route completion, with Wilson 95 % intervals. Validity of each certificate is assessed over 200 random resplits of the pooled calibration and test scenarios. A valid certificate keeps the frequency of test miss rate above α at or below δ . Harm is scored with the single nuPlan definition above; its sensitivity to the time-to-collision threshold is not evaluated here (Section VI).

G. Pre-registration and reproducibility

The harm definition, $\alpha = 0.05$, $\delta = 0.10$, and the refutation criteria for three hypotheses were recorded in a version-controlled note before any calibration or test data were collected: **H1**, that open-loop metrics rank triggers differently from closed-loop outcomes; **H2**, that a certified trigger achieves a better safety-conservatism trade-off than a tuned fixed threshold; and **H3**, that a threshold certified on Argoverse 2 loses its guarantee on nuScenes. The note carries dated amendments (a corrected confidence baseline, the exactness measurements, a Waymo target and a reverse-shift analysis), each written before the corresponding outcomes were evaluated, and it labels the analyses run afterwards as post hoc. Code, per-scenario rollout records and configuration files will be released with the paper. Each scenario's record contains every score trace and outcome, so every table can be regenerated without re-simulation.

V. RESULTS

All results use $\alpha = 0.05$ and $\delta = 0.10$, 1,994 calibration and 1,996 test scenarios from Argoverse 2, and 1,504 nuScenes scenarios, with Wilson 95 % intervals unless stated. Dropping the 3.6 % of scenarios with any flagged simulator drift changes no conclusion below. Sections V-A–V-D are the pre-registered analyses. The three subsections that follow them (the consequences of the intervention and the comparison with a class-conditional certificate, the reverse shift, and few-label recalibration) are post hoc or partly so, and are labelled.

A. Does a certificate keep its promise, and what does it cost? (RQ1)

Table II gives every method at its operating point on the Argoverse 2 test set. The certified trigger (LTT) misses 3.7 % of harmful events [95 % CI 2.9–4.6], below the 5 % target, with 28.5 % unnecessary stops and 2.6 % fallback-induced collisions. The last column repeats the calibration on 200 random resplits and reports how often the realised test miss rate exceeds α . A valid certificate keeps this at or below $\delta = 10$ %. LTT does (9.5 %). The same geometric score with a threshold tuned empirically, and its Conformal Risk Control (CRC) counterpart, which guarantees only an expectation, exceed the target in 43.5 % and 41.0 % of resplits, and the tuned confidence trigger in 50.5 %. Certification is therefore the only design here that keeps its target reliably.

The certificate has a price, and it does not win the trade-off. The tuned geometric threshold stops less often than LTT (26.0 % versus 28.5 %; paired bootstrap difference -2.5 points, 95 % CI $[-3.2, -1.8]$), and CDT stops less still (24.1 %). Our pre-registered hypothesis H2 required no baseline to be within 3 points of the certified trigger; it is **refuted**. The supportable claim is narrower: among the designs whose promise can be checked, the certified one is the only one that reliably meets it, and it pays about 2.5 points of unnecessary stops for that.

Deployed-practice triggers are far more conservative. Tuned to the same miss target, the confidence and MC-dropout triggers stop in 68.3 % and 70.8 % of scenarios, 40–45 points

TABLE II
CLOSED-LOOP OUTCOMES ON THE ARGOVERSE 2 TEST SET AT $\alpha = 0.05$.
RATES IN %. LAST COLUMN: FRACTION OF 200 CALIBRATION/TEST
RESPLITS IN WHICH THE TEST MISS RATE EXCEEDS α (VALID IF ≤ 10).

Method	Miss	Unnec.	Induced	Auton.	Viol.
No fallback	15.1	0.0	0.0	100	–
Always fire	0.0	84.9	6.5	0	–
T1 confidence	4.6	68.3	4.0	20.3	50.5
T2 MC-dropout spread	4.0	70.8	4.9	17.6	–
T3 open-loop conformal	0.0	73.8	5.5	11.1	0.0
Tuned geometric (no cert.)	4.3	26.0	2.3	61.4	43.5
CRC (geometric)	4.3	26.0	2.3	61.4	41.0
LTT (geometric, certified)	3.7	28.5	2.6	58.5	9.5
CDT (online)	4.9	24.1	2.2	63.9	–
ACI (online)	1.3	45.0	3.5	40.2	–
Oracle (hindsight)	5.0	23.6	2.2	64.2	–

TABLE III
OPEN-LOOP AUROC VERSUS CLOSED-LOOP UNNECESSARY STOPS AT
THE CERTIFIED THRESHOLD, ARGOVERSE 2 TEST.

Score	AUROC	Unnec. stops (%)
Geometric, isotropic radius	0.843	27.5
Geometric, length-aware box	0.832	28.5
Predictor gap	0.817	31.8
Geometric, route corridor	0.577	84.9
Confidence	0.448	75.8
Ensemble spread	0.442	72.3

more than the certified geometric trigger (paired CIs exclude zero by a wide margin), and the open-loop conformal trigger reaches a miss rate of 0.0 % only by firing in 73.8 %. Its certificate is valid in the sense that it never violates, and useless in the sense that a vehicle that stops in three of four scenarios has abandoned its task. ACI reaches 1.3 % misses at 45 % stops, which is consistent with a guarantee on the prediction not translating into a useful decision.

B. Do open-loop metrics choose the closed-loop trigger? (RQ2)

Table III ranks six trigger scores by open-loop AUROC of their scenario peak for predicting harm, and by closed-loop unnecessary stops at each score's certified threshold. Rankings agree at the top and diverge below it: Kendall's $\tau = 0.60$ on Argoverse 2 and 0.47 on nuScenes, and the top-ranked score is the same on Argoverse 2 and differs on nuScenes only by 0.2 points of stops. The pre-registered rule refutes H1 only if $\tau \geq 0.8$ and the top score coincides in every setting, so H1 is not refuted, but the evidence is weak: open-loop AUROC picked the closed-loop best score, or a statistical tie, in every setting. The clearest disagreement is at the bottom. The confidence score has AUROC 0.45, below chance, meaning higher confidence-based uncertainty is on average associated with *less* harm, yet a threshold on it can still reach the miss target, at 75.8 % stops (certified) rather than by discriminating harmful from safe scenarios.

C. Ablations and latency (RQ4)

Table IV shows that two components matter and one does not. Using the vehicle's planned, route-following path

TABLE IV
ABLATIONS ON THE ARGOVERSE 2 TEST SET, CERTIFIED BY LTT.

Variant	Miss (%)	Unnec. stops (%)
Full score, LTT, no latency	3.7	28.5
Route-following ego corridor	0.0	84.9
Width-only isotropic radius	3.6	27.5
Confidence instead of geometry	3.6	75.8
Ensemble spread instead of geometry	3.6	72.3
CRC instead of LTT	4.3	26.0
Empirical tuning instead of LTT	4.3	26.0
+1 decision tick latency (0.5 s)	5.3	26.3
+2 decision ticks latency (1.0 s)	9.1	23.4

TABLE V
A THRESHOLD CERTIFIED ON ARGOVERSE 2 (LTT, $\alpha = 0.05$) DEPLOYED ON nuSCENES. “AVOIDABLE” EXCLUDES THE 5.0 % OF nuSCENES SCENARIOS WITH HARM AT THE FIRST DECISION TICK (POST HOC).

Setting	Miss (%) [95 % CI]	Unnec. (%)	n
Argoverse 2 test (in domain)	3.7 [2.9, 4.6]	28.5	1,996
nuScenes, unweighted	13.2 [11.6, 15.0]	26.1	1,504
nuScenes, weighted (no labels)	7.0 [5.8, 8.4]	52.4	1,504
nuScenes, oracle threshold	5.0	77.9	1,504
Avoidable, unweighted	9.2 [7.8, 10.8]	27.1	1,429
Avoidable, weighted ($n_{\text{eff}} = 68$)	2.2 [1.6, 3.1]	54.5	1,429

instead of a kinematic corridor makes the trigger fire in essentially every scenario (84.9 % stops), consistent with the route reaching conflict points that the harm label, which uses a kinematic extrapolation, does not count. Replacing geometry with confidence or ensemble spread costs 44–47 points of stops. The length-aware box does not matter: a width-only isotropic radius performs the same (27.5 % versus 28.5 %). Delaying the intervention by one decision tick (0.5 s) already lifts the miss rate above the target (5.3 %) and two ticks give 9.1 %, while stops fall, so a certificate calibrated at zero latency is not valid for a slower actuation chain and latency belongs in the certified quantity.

D. Dataset shift (RQ3)

The Argoverse 2 certificate does not transfer. On nuScenes the threshold certified at a 5 % target misses 13.2 % [11.6, 15.0], 2.6 times the target and with an interval that excludes it; the pre-registered hypothesis H3 is **supported** (15.2 % after dropping drift scenarios). Part of this is a floor rather than shift: 5.0 % of nuScenes scenarios already contain harm at the first decision tick, which even a trigger that always fires cannot prevent (0.1 % on Argoverse 2). That floor equals the target, so no threshold can be certified at $\alpha = 0.05$ on nuScenes. With those scenarios excluded (post hoc), the shifted miss rate is 9.2 % [7.8, 10.8], still above the target.

Density-ratio weighting, using only unlabelled target covariates, moves the miss rate from 13.2 % to 7.0 %, still above the target, and doubles unnecessary stops (26.1 % to 52.4 %). On the avoidable subset it reaches 2.2 %, below the target, but its effective sample size is only 68, so the certificate is achieved almost entirely through conservatism (54.5 % stops).

TABLE VI
OUTCOME PARTITION (% OF SCENARIOS). RESIDUAL = MISS + TOO LATE + NEW HARM. INDUCED: NOT-AT-FAULT COLLISION THE NO-FALLBACK RUN AVOIDS.

Data	Trigger	Miss	Too late	Prev.	New harm	Resid. / Induced
AV2	No fallback	15.1	0.0	0.0	0.0	15.1 / 0.0
AV2	LTT (certified)	3.7	1.7	9.8	1.1	6.5 / 2.6
AV2	Tuned geometric	4.3	1.7	9.1	0.9	6.9 / 2.3
AV2	Always fire	0.0	1.0	14.1	3.1	4.1 / 6.5
nuScenes	LTT (AV2-cert.)	13.2	1.0	8.2	1.7	15.6 / 2.4
nuScenes	Always fire	5.0	1.6	16.2	3.8	9.7 / 6.4

E. What the intervention does, and a class-conditional certificate (post hoc)

The certified quantity is the alert. Table VI partitions every scenario by what happens after the fallback fires. At the certified threshold on Argoverse 2, 3.7 % of scenarios are missed, in a further 1.7 % the fallback fires before harm but harm still occurs, and in 9.8 % harm is prevented. In a further 1.1 % of all scenarios, which are harmless without a fallback, the intervened run contains harm (nuPlan definition), and 2.6 % suffer an induced collision. Harm in the intervened run is therefore 6.5 % (15.1 % without a fallback), almost twice the certified miss rate. A trigger that always fires creates harm in 3.1 % of scenarios. On nuScenes the same threshold leaves 15.6 % residual harm. The fallback removes a large share of harm, but the certificate does not speak to what remains.

Reading a 5 % marginal target. Harm occurs in 15.8 % of calibration scenarios, so the certified 3.7 % miss rate leaves 24 % of harmful scenarios (58 % on nuScenes) without an alert before harm.

Class-conditional certification. Table VII implements the Luo *et al.* certificate in our framework: Mondrian split conformal prediction on the unsafe class, where unsafe means harm in the reference run and an alert means firing before it, with the finite-sample correction and without their tie randomisation. At $\epsilon = 0.05$ on the 316 unsafe calibration scenarios it selects a threshold equivalent to always firing, so the class-conditional target is vacuous for this predictor at this level: the score separates harmful from harmless scenarios well enough for a marginal target (AUROC 0.83) but not for a 5 % miss rate among harmful ones. At the marginal-matched $\epsilon = \alpha/\pi = 0.32$ it selects the same threshold as empirical tuning. Deployed on nuScenes, that certificate has a conditional miss rate of 66 % against its 32 % target. The harm base rate rises from 15.8 % to 22.1 %, but the conditional miss rate is far above what that alone explains, so the class-conditional guarantee does not survive the dataset shift either.

F. Reverse shift (exploratory, post hoc)

We also certified on half of the nuScenes scenarios and deployed on Argoverse 2 test, a direction pre-registered as a reverse-shift analysis. As pre-registered, with all scenarios, the procedure *refused to certify* on both halves, because of the 5.0 % floor. On the avoidable subset (post hoc), thresholds certified on nuScenes ($n \approx 715$) give an Argoverse 2 miss rate of 1.9 % and 2.0 %, both below the target, at 40.6 % and 39.5 % unnecessary stops. A certificate from the harder dataset

TABLE VII

MARGINAL CERTIFICATE (LTT) VERSUS CLASS-CONDITIONAL CERTIFICATES (LUO-STYLE), THRESHOLDS CERTIFIED ON ARGOVERSE 2. MISS: MARGINAL MISS RATE; COND.: MISS RATE AMONG SCENARIOS WITH HARM IN THE REFERENCE RUN.

Certificate	Argoverse 2 test			nuScenes		
	Miss	Cond.	Unnec.	Miss	Cond.	Unnec.
LTT, marginal $\alpha = 0.05$	3.7	24.2	28.5	13.2	58.1	26.1
Luo-style, $\epsilon = 0.05$	0.0	0.0	84.9	5.0	19.6	77.9
Luo-style, $\epsilon = 0.32$	4.3	28.5	26.0	14.9	66.0	21.3

TABLE VIII

EXPLORATORY: RECALIBRATION WITH k LABELLED nuSCENES SCENES (AVOIDABLE SUBSET, 60 RANDOM DRAWS). VIOL.: FRACTION OF DRAWS WITH TEST MISS RATE ABOVE 5 %.

k	Method	Miss (%)	Viol. (%)	Unnec. (%)
0	AV2-certified	9.2	–	27.1
40	Target-only tuned	6.6	67	35.6
40	Target-only LTT	0.0	0	81.3
40	AV2 + k weighted LTT	1.8	10	63.6
60	Target-only tuned	6.0	70	36.7
60	Target-only LTT	0.8	0	72.5
60	AV2 + k weighted LTT	2.8	0	53.3

therefore transfers safely but conservatively, while the reverse does not transfer. This asymmetry is a single-pair observation pending replication.

G. Recalibration with a few labelled target scenes (exploratory)

Table VIII draws k labelled nuScenes scenes at random (60 draws each) and evaluates on the rest, on the avoidable subset. This analysis was not pre-registered. Uncertified tuning on 40–60 target scenes exceeds the target in about two draws in three. A certified threshold on the same scenes keeps the target (violations 0–10 %) but stops in 53–81 % of scenarios, against 27 % without target labels. A handful of labelled scenes therefore restores a *valid* certificate, not the original operating point.

H. Second shifted dataset and sensitivity

Waymo Open Motion. [Pending: miss rate of the Argoverse 2-certified threshold on Waymo (pre-registered as H3-W: the 95 % interval excludes α), weighted repair, and open-loop versus closed-loop ranking.]

Sensitivity. [Pending: non-reactive log-replay traffic, a 2.0 m/s² comfort MRM, and 10 Hz decisions on 100 scenarios.]

VI. SCOPE AND FUTURE WORK

The claims above are deliberately scoped. Each boundary below is a design choice with a stated rationale, and each suggests a direct extension.

What we claim, and what we do not. We claim that certification is what makes a closed-loop trigger keep its stated miss rate, that it costs a measurable number of unnecessary stops, and that the guarantee is fragile to dataset shift and to

latency. We do not claim a better-performing trigger: a tuned threshold and an online method stop less often, and we report the pre-registered hypothesis that said otherwise as refuted.

One predictor, one simulator. All results use one trajectory predictor trained on Argoverse 2, and MetaDrive with rule-based (IDM) agents. Whether the ordering of triggers survives a stronger predictor or learned agent behaviour is open. The protocol accepts both unchanged, and the geometric-versus-confidence gap is large enough that we expect it to persist, which we have not tested.

Unavoidable harm. The 5.0 % first-tick floor on nuScenes shows that a miss metric defined on harm anywhere in a replayed snippet mixes events a fallback could prevent with events already under way. We report the pre-registered definition and the avoidable-only variant side by side, and propose the latter as a refinement for future benchmarks.

Marginal versus class-conditional targets, and alert versus outcome. The certificate bounds the probability, over the scenario distribution, that harm is not pre-empted by an alert. That is the quantity a deployment authority can audit, but it is a weaker statement than a bound among harmful scenarios, and it bounds the alert, not the outcome: harm still occurs after some alerts and is sometimes created by the intervention. Certifying outcome-level harm, or a class-conditional miss rate, would need a stronger score or a different loss. It does not bound the risk of any individual scenario, and stratified certification by scenario type follows from the same procedure applied within strata.

Covariate shift and few-shot recalibration. The weighted certificate covers shift in observable scenario covariates, and its effective sample size, 68 in our nuScenes subset, shows when it becomes unreliable. The few-label recalibration result is exploratory and single-pair; it needs replication on a second target before being relied on.

Harm definition. Harm is scored with the nuPlan definition alone. Sensitivity to the time-to-collision threshold and to fault attribution is a natural robustness study, and a severity-weighted or tail (CVaR) certificate in the sense of [9] requires only a change of loss.

Object states as input; one fallback family. The predictor receives agent states, isolating the triggering decision from perception error, and we evaluate stop-in-lane manoeuvres at the UN R157 deceleration limit. Composing the certificate with a calibrated perception module, and other fallbacks such as a lateral pull-over or hand-over, change the quantities priced but not the method.

VII. CONCLUSION

A safety guarantee on a fallback trigger is not a permanent property of the model; it depends on the data it was certified on, the timing of the intervention, and what the fallback itself does. Evaluated exactly with the fallback executed, a Learn-then-Test-certified trigger keeps its 5 % target on the calibration distribution (3.7 % missed, violations in 9.5 % of resplits against 43.5 % without a certificate) at a cost of about 2.5 points of unnecessary stops, and deployed-practice confidence and ensemble triggers need 68–71 % stops for similar

safety. The certificate controls the alert, not the outcome: harm in the intervened run is 6.5 %, 1.1 % of it created by the intervention, and a 5 % marginal target leaves about a quarter of harmful scenarios unalerted. A class-conditional certificate in the style of Luo *et al.* is vacuous at 5 % for this predictor. The same certificate fails on nuScenes (13.2 % missed), is broken by one decision tick of latency, and can be unreachable when the target data contain harm no trigger can prevent. Recalibrating with a few dozen labelled target scenes restores a valid certificate at a large operating cost in an exploratory analysis. [Waymo replication to be added.] These findings argue for treating certification, shift monitoring and latency as one audited quantity, and the evaluation protocol makes that audit inexpensive to repeat for any trigger.

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REFERENCES

- [1] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*. Berlin, Germany: Springer-Verlag, 2005.
- [2] A. N. Angelopoulos, S. Bates, E. J. Candès, M. I. Jordan, and L. Lei, “Learn then test: Calibrating predictive algorithms to achieve risk control,” *Ann. Appl. Statist.*, vol. 19, no. 2, pp. 1641–1662, 2025.
- [3] A. N. Angelopoulos, S. Bates, A. Fisch, L. Lei, and T. Schuster, “Conformal risk control,” *arXiv preprint arXiv:2208.02814*, 2022.
- [4] K. Rahaman, J. V. Deshmukh, A. R. Hota, and L. Lindemann, “When environments shift: Safe planning with generative priors and robust conformal prediction,” in *Proc. 8th Annu. Conf. Learn. Dyn. Control (LADC)*, 2026.
- [5] A. Doula, M. Mühlhäuser, and A. S. Guinea, “SafePath: Conformal prediction for safe LLM-based autonomous navigation,” *arXiv preprint arXiv:2505.09427*, 2025.
- [6] A. Farid, S. Veer, B. Ivanovic, K. Leung, and M. Pavone, “Task-relevant failure detection for trajectory predictors in autonomous vehicles,” in *Proc. 6th Conf. Robot Learn. (CoRL)*, Auckland, New Zealand, 2022.
- [7] S. Ettinger *et al.*, “Large scale interactive motion forecasting for autonomous driving: The Waymo Open Motion Dataset,” in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2021, pp. 9710–9719.
- [8] R. Luo, S. Zhao, J. Kuck, B. Ivanovic, S. Savarese, E. Schmerling, and M. Pavone, “Sample-efficient safety assurances using conformal prediction,” *Int. J. Robot. Res.*, vol. 43, no. 9, pp. 1409–1424, 2024.
- [9] M. P. Chapman, R. Bonalli, K. M. Smith, I. Yang, M. Pavone, and C. J. Tomlin, “Risk-sensitive safety analysis using conditional value-at-risk,” *IEEE Trans. Autom. Control*, vol. 67, no. 12, pp. 6521–6536, Dec. 2022.
- [10] Q. Li, Z. Peng, L. Feng, Z. Liu, C. Duan, W. Mo, and B. Zhou, “ScenarioNet: Open-source platform for large-scale traffic scenario simulation and modeling,” in *Proc. 37th Conf. Neural Inf. Process. Syst. (NeurIPS), Track Datasets and Benchmarks*, 2023.
- [11] N. Karnchanachari *et al.*, “Towards learning-based planning: The nuPlan benchmark for real-world autonomous driving,” in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2024.
- [12] J. Lekeufack, A. N. Angelopoulos, A. Bajcsy, M. I. Jordan, and J. Malik, “Conformal decision theory: Safe autonomous decisions from imperfect predictions,” in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2024, pp. 11668–11675.
- [13] I. Gibbs and E. Candès, “Adaptive conformal inference under distribution shift,” in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 34, 2021, pp. 1660–1672.
- [14] R. J. Tibshirani, R. Foygel Barber, E. J. Candès, and A. Ramdas, “Conformal prediction under covariate shift,” in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 32, 2019.
- [15] Y. Gal and Z. Ghahramani, “Dropout as a Bayesian approximation: Representing model uncertainty in deep learning,” in *Proc. 33rd Int. Conf. Mach. Learn. (ICML)*, 2016, pp. 1050–1059.
- [16] R. Girgis, F. Golemo, F. Codevilla, M. Weiss, J. A. D’Souza, S. E. Kahou, F. Heide, and C. Pal, “Latent variable sequential set transformers for joint multi-agent motion prediction,” in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2022.
- [17] L. Feng, M. Bahari, K. M. Ben Amor, É. Zablocki, M. Cord, and A. Alahi, “UniTraj: A unified framework for scalable vehicle trajectory prediction,” in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2024.
- [18] United Nations Economic Commission for Europe, “UN Regulation No. 157 — Uniform provisions concerning the approval of vehicles with regard to Automated Lane Keeping Systems,” Rev. 1, 2025.
- [19] R. Sinha, E. Schmerling, and M. Pavone, “Closing the loop on runtime monitors with fallback-safe MPC,” in *Proc. IEEE Conf. Decis. Control (CDC)*, 2023.
- [20] S. Sohrabi *et al.*, “Quantifying the automated vehicle safety performance: A scoping review of the literature, evaluation of methods, and directions for future research,” *Accid. Anal. Prev.*, vol. 152, Art. no. 106003, 2021.
- [21] D. Xie *et al.*, “A survey on scenario-based test and evaluation methods for autonomous vehicles,” *Proc. Inst. Mech. Eng. D, J. Automobile Eng.*, 2025.